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Steering of roaming in wireless communication networks

Abstract

A method for steering of roaming (SOR) in wireless communication networks includes sending by a mobile equipment (ME) a registration message to a visited public land mobile network (VPLMN). The ME receives a first message from a home public land mobile network (HPLMN) through the VPLMN. The ME determines that the first message has failed a security check. In response to determining that the first message has failed the security check, the ME sends a second message to the VPLMN, the second message including an indication that the first message has failed the security check.

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References Cited

U.S. PATENT DOCUMENTS

Patent No.	Issued Date	Patentee Name	U.S. Cl.	CPC
10952062	12/2020	Buckley et al.	N/A	N/A
2010/0064344	12/2009	Wang	N/A	N/A
2010/0322216	12/2009	Roger et al.	N/A	N/A
2012/0282898	12/2011	Lee et al.	N/A	N/A
2013/0109377	12/2012	Al-Khudairi et al.	N/A	N/A
2013/0217391	12/2012	Klatt	455/435.2	H04W 8/24
2014/0357315	12/2013	John	N/A	N/A
2015/0065125	12/2014	Patel et al.	N/A	N/A
2018/0227872	12/2017	Li et al.	N/A	N/A
2018/0332554	12/2017	Kawasaki	N/A	H04W 80/04
2019/0268752	12/2018	Buckley et al.	N/A	N/A
2021/0185508	12/2020	Buckley et al.	N/A	N/A

FOREIGN PATENT DOCUMENTS

Patent No.	Application Date	Country	CPC
2011199390	12/2010	JP	N/A
2019017689	12/2018	WO	N/A

OTHER PUBLICATIONS

Samsung, p-CR for securely transmitting Preferred PLMN list to UE, Nov. 27-Dec. 1, 2017, 3GPP TSG-CT WG1 Meeting #107, C1-175066, Reno (USA). (Year: 2017). cited by examiner

Samsung, Ericsson , Updates to Stage 2 solution of Steering Of Roaming (SOR), Date: Apr. 16, 2018, 3GPP TSG-CT WG1 Meeting #110, C1-182338. (Year: 2018). cited by examiner

3rd Generation Partnership Project, "3GPP TS 23.501 v15.0.0: 3rd Generation Partnership Project; Technical Specification Group Services and Systems Aspects; System Architecture for the 5G System; Stage 2 (Release 15)," 3GPP 23.501, Dec. 2017, 181 pages. cited by applicant

3rd Generation Partnership Project, "3GPP TS 23.502 v15.0.0: 3rd Generation Partnership Project; Technical Specification Group Services and Systems Aspects; Procedures for the 5G System; Stage 2 (Release 15)," 3GPP TS 23.502, Dec. 2017, 258 pages. cited by applicant

3rd Generation Partnership Project, "3GPP TS 33.501 v0.6.0: 3rd Generation Partnership Project;

Technical Specification Group Services and System Aspects; Security architecture and procedures for 5G System (Release 15),” 3GPP TS 33.501, Dec. 2017, 79 pages. cited by applicant

Aboba et al., “Extensible Authentication Protocol (EAP),” RFC 3748, the Internet Society, Jun. 2004, 67 pages. cited by applicant

Advisory Action issued in United States U.S. Appl. No. 15/952,846, filed Apr. 22, 2019, 2 pages. cited by applicant

Arkko et al., “Extensible Authentication Protocol Method for 3rd Generation Authentication and Key Agreement (EAP-AKA),” RFC 4187, the Internet Society, Jan. 2006, 79 pages. cited by applicant

Arkko et al., “Improved Extensible Authentication Protocol Method for 3rd Generation Authentication and Key Agreement (EAP-AKA),” RFC 5448, IETF Trust, May 2009, 29 pages. cited by applicant

Communication Pursuant to Article 94 (3) EPC issued in European Application No. 19710543.0, dated Sep. 15, 2021, 12 pages. cited by applicant

European Telecommunications Standards Institute, “ETSI TS 102 223 V14.0.0, Technical Specification: Smart Cards; Card Application Toolkit (CAT) (Release 14),” ETSI TS 102 223, May 2017, 247 pages. cited by applicant

European Telecommunications Standards Institute, “ETSI TS 102 225 v12. 1.0, Technical Specification, Smart Cards; Secured packet structure for UICC based applications (Release 12),” ETSI TS 102 225 V12.1.0, Oct. 2014, 22 pages. cited by applicant

Final Office Action issued in United States U.S. Appl. No. 15/952,846, filed Dec. 6, 2019, 21 pages. cited by applicant

Final Office Action issued in United States U.S. Appl. No. 15/952,846 on Feb. 14, 2019, 20 pages. cited by applicant

Huawei et al., “Discussion Paper for Steering of Roaming” 3GPP Draft, Discussion Paper For Steering of Roaming, 3GPP TSG SA WG3 (Security), Meeting #89, S3-173235, Sophia-Antipolis Cedex, France, Nov. 27-Dec. 1, 2017, 5 pages. cited by applicant

Kaufman et al., “Internet Key Exchange Protocol Version 2 (IKEv2),” RFC 7269, Internet Engineering Task Force (IETF), Oct. 2014, 142 pages. cited by applicant

Non-Final Office Action issued in U.S. Appl. No. 15/952,846, filed Jul. 8, 2019, 25 pages. cited by applicant

Non-Final Office Action issued in U.S. Appl. No. 15/952,846, filed Sep. 25, 2018, 27 pages. cited by applicant

Notice of Allowance issued in U.S. Appl. No. 15/952,846, filed Oct. 16, 2020, 9 pages. cited by applicant

Notice of Allowance issued in U.S. Appl. No. 17/185,761, filed May 5, 2022, 17 pages. cited by applicant

Office Action for Chinese Patent Application No. 201980028204.7, Jul. 15, 2023, 16 Pages. cited by applicant

Office Action in Japanese Appln. No. 2020-544770, dated Jan. 25, 2023, 10 pages (with English translation). cited by applicant

Official Action for Japanese Patent Application No. 2020-544770, dated Jul. 3, 2023, 6 Pages. cited by applicant

PCT International Search Report & Written Opinion issued in International Application No. PCT/US2019/019508, dated Apr. 10, 2019, 20 pages. cited by applicant

Qualcomm Incorporated, “Control Plane Solution for Steering of Roaming in SGS,” 3GPP TSG-CT WGI Meeting #106 (cl-174151), Kochi (India), Oct. 23-27, 2017, 3 pages. cited by applicant

Samsung et al., “Stage 2 Solution of Steering of Roaming (SOR) based on Authentication Procedure during 2Registration (Alternative 3)” Change Request, C1-181108, 3GPP TTSG-CT WG1, Meeting #109, Montreal, Canada, Feb. 26-Mar. 2, 2018, 18 pages. cited by applicant

Samsung et al., “Updates to Stage 2 Solution of Steering of Roaming” Change Request, C1-182338, 3GPP TSG-CT, Meeting #110, Kunming, PR of China, Apr. 16-20, 2018, 24 pages. cited by applicant

Samsung, “p-CR for Securely Transmitting Preferred PLMN List to UE,” 3GPP TSG-CT WGI Meeting #107 (CI-175066), Reno (USA), Nov. 27-Dec. 1, 2017, 4 pages. cited by applicant

“3 GPP TS 24.008 v 15. 1.0: 3rd Generation Partnership Project; Technical Specification Group Core Network and Terminals; Mobile Radio Interface Layer 3 specification; Core network protocols; Stage 3 (Release 15)”, 3rd Generation Partnership Project, 3GPP TS 24.008, Dec. 2017, 785 pages. cited by applicant

“3GPP TS 23.122 V15.2.0, 3rd Generation Partnership Project; Technical Specification Group Core Network and Terminals; Non-Access-Stratum (NAS) function related to Mobile Station (MS) in idle mode (Release 15),”, 3rd Generation Partnership Project, 3GPP TS 23.122, Dec. 2017, 54 pages. cited by applicant

“3GPP TS 24.302 V15.1.1: 3rd Generation Partnership Project; Technical Specification Group Core Network and Terminals; Access to the 3GPP Evolved Packet Core (EPC) via non-3GPP access networks; Stage 3 (Release 15),”, 3rd Generation Partnership Project, 3GPP TS 24.302, Dec. 2017, 171 pages. cited by applicant

“3GPP TS 24.501 V0.2.2: 3rd Generation Partnership Project; Technical Specification Group Core Network and Terminals; Non-Access-Stratum (NAS) Protocol for 5G System (SGS); Stage 3 (Release 15),”, 3rd Generation Partnership Project, 3GPP TS 24.501, Dec. 2017, 179 pages. cited by applicant

“3GPP TS 31.111 v15.0.0: 3rd Generation Partnership Project; Technical Group Core Network and Terminals; Universal Subscriber Identity Module (USIM) Application Toolkit (USAT) (Release 15),”, 3rd Generation Partnership Project, 3GPP TS 31.111, Dec. 2017, 148 pages. cited by applicant

“ETSI TS 123 122 V14.4.0, Technical Specification, Digital cellular telecommunications system (Phase 2+) (GSM); Universal Mobile Telecommunications System (UMTS);”, 3rd Generation Partnership Project/European Telecommunications Standards Institute, LTE; Non-Access-Stratum (NAS) functions related to Mobile Station (MS) in idle mode, ETSI TS 123 122/3GPP TS 23.122, Oct. 2017, 57 pages. cited by applicant

“ETSI TS 131 102 V14.4.0, Technical Specification, Universal Mobile Telecommunications System (UMTS)”, 3rd Generation Partnership Project/European Telecommunications Standards Institute, LTE; Characteristics of the Universal Subscriber Identity Module (USIM) application (3 GPP TS 31,102 version 14.4.0 Release 14), Jan. 2018, 298 pages. cited by applicant

“ETSI TS 131 111 V10.4.0, Technical Specification, Digital cellular telecommunications system (Phase 2+); Universal Mobile Telecommunications System (UMTS); LTE; Universal Subscriber Identity Module (USIM) Application Toolkit (USAT)”, 3rd Generation Partnership Project/European Telecommunications Standards Institute, (3GPP Ts 31.111 version 10.4.0 Release 10, ETSI TS 131 111/3GPP TS 31.111, Nov. 2011, 126 pages. cited by applicant

Samsung et al., “Updates to Stage 2 Solution of Steering of Roaming (SOR)” Change Request, C1-182338, 3GPP Draft, Apr. 9, 2018, 9 Pages. cited by applicant

The Extended European Search Report for European Patent Application No. 24156289.1, Sep. 19, 2024, 26 Pages. cited by applicant

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Background/Summary

CROSS REFERENCE TO RELATED APPLICATIONS (1) This application is a continuation of U.S. application Ser. No. 17/882,356, filed Aug. 5, 2022, which is a continuation of U.S. application Ser. No. 17/185,761, filed Feb. 25, 2021, now issued as U.S. Pat. No. 11,438,755, which is a continuation of U.S. application Ser. No. 15/952,846, filed Apr. 13, 2018, now issued as U.S. Pat. No. 10,952,062, which claims priority to U.S. Provisional Application No. 62/635,483, filed Feb. 26, 2018, the entire contents of which are hereby expressly incorporated by reference herein in its entirety.

FIELD OF THE INVENTION

(1) This disclosure relates to roaming in wireless communication networks.

BACKGROUND

(2) Steering of user equipment (UE) in a visited public land mobile network (VPLMN) is a procedure enabling a home public land mobile network (HPLMN) to update a list of preferred PLMN/access technology combinations at the UE via non-access stratum (NAS) signaling. The HPLMN updates the list of preferred PLMN/access technology combinations, e.g., depending on the public land mobile network (PLMN) where the UE is registered or when required by HPLMN operator policies. Steering of UE in VPLMN may also be known as steering of roaming (SOR). SOR enables a HPLMN to steer a user equipment (UE) from one network to another. SOR is a technique whereby a roaming UE is encouraged to roam to a preferred roamed-to network by the HPLMN. For example, a UE is registered on one public land mobile network (PLMN), and for some reason the UE's HPLMN wants the UE to register on another PLMN.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) FIG. 1 is a data flow diagram illustrating an example process of using an authentication procedure to transport a list of preferred public land mobile network (PLMN) and access technology combinations.

(2) FIG. 2 is a data flow diagram illustrating an example steering of roaming (SOR) procedure, according to some implementations of the present disclosure.

(3) FIGS. 3A-3B illustrate a data flow diagram of an example registration procedure from 3GPP TS 23.502.

(4) FIGS. 4A-4C illustrate an example description for including a device type indicator in a REGISTRATION REQUEST message, according to some implementations of the present disclosure.

(5) FIG. 5 illustrates an example description for including a device type indicator in an Extensible Authentication Protocol (EAP) message, according to some implementations of the present disclosure.

(6) FIG. 6 is a data flow diagram illustrating an example procedure using EAP signaling to send SOR data, according to some implementations of the present disclosure.

(7) FIG. 7 is a data flow diagram illustrating an example process using EAP signaling for obtaining SOR data in fifth generation (5G) networks, according to some implementations of the present disclosure.

(8) FIGS. 8A-8E illustrate an example description of sending SOR data in EAP-Authentication and Key Agreement (AKA), according to some implementations of the present disclosure.

(9) FIG. 9 illustrates an example description of a REFRESH command, according to some implementations of the present disclosure.

(10) FIG. 10 illustrates an example description of an environment file (EF) for data “Operator controlled PLMN selector with Access Technology”, according to some implementations of the present disclosure.

(11) FIG. 11 is a data flow diagram illustrating an example SOR procedure, according to some implementations of the present disclosure.

(12) FIGS. 12A-12B illustrate an example description for the SOR procedure of FIG. 11, according to some implementations of the present disclosure.

(13) FIG. 13 is a schematic illustrating an example network node according to some implementations of the present disclosure.

(14) FIG. 14 is a schematic illustrating an example UE apparatus according to some implementations of the present disclosure.

(15) FIGS. 15A-15F illustrate an example description for steering of UE in visited PLMN (VPLMN) during registration and after registration, according to some implementations of the present disclosure.

(16) Like reference numbers and designations in the various drawings indicate like elements.
DETAILED DESCRIPTION

(17) The present disclosure is directed to steering of roaming (SOR) in wireless communication networks (i.e. steering of user equipment (UE) in visited public land mobile network (VPLMN)). In some wireless networks, such as fifth generation (5G) networks, a home operator (e.g., home public land mobile network (HPLMN)) can steer a UE from one network to another. For example, a UE is registered on one public land mobile network (PLMN), and the UE's HPLMN may want the UE to register on another PLMN.

(18) A UE can perform a PLMN search to find an alternative PLMN. For example, when the UE first powers up, the UE can perform an initial PLMN search. After the UE powers up, the UE can perform the PLMN search periodically. The periodical PLMN search, for example, enables the UE to find a PLMN which has a higher priority than the UE's current PLMN (the PLMN that the UE is currently registered is also called the UE's VPLMN or Registered PLMN (RPLMN)). For example, the UE is on a VPLMN (2nd network) other than its HPLMN (1st network), and the UE can periodically search for the HPLMN. The periodical PLMN search can happen when a timer expires. An example of such a timer is a timer known as timer T. The PLMN search may only take place when the UE is in an IDLE mode, IDLE state, 5GMM-IDLE mode, or CM-IDLE state e.g. a state where there is no active communications taking place with the network. If the UE performed a PLMN search/selection in a CONNECTED mode, any connection would be lost as the radio would have to disconnect from the current PLMN. Therefore, if the timer T expires, the UE has to wait until the UE is in an idle mode to perform the PLMN search.

(19) In some cases, such as in fourth generation (4G), third generation (3G), or second generation (2G) systems, once a UE has attached to a VPLMN, the HPLMN sends a Short Message (SM) including a secured packet (as defined in ETSI TS 102 225). This secured packet includes at least one PLMN identity. The SM is received by the Mobile Equipment (ME) of the UE, and based on an indication (e.g., code point) in the SM the ME determines that some or all of the contents of the SM is for the universal integrated circuit card (UICC) of the UE. The UICC then receives the SM and unpacks the SM to determine that it includes a secured packet. The content of the secured packet updates the Preferred Operator PLMN list (e.g., the most top entry) in the UICC, and the UICC sends a Universal Subscriber Identity Module (USIM) Application Toolkit (USAT) REFRESH command to the ME. Upon receipt of the REFRESH command, the ME reads the Preferred Operator PLMN list from the UICC or from the REFRESH command if a PLMN list was included, and stores the list in the memory of the ME. The ME then performs a PLMN search taking into account the PLMNs in the updated Preferred Operator PLMN list. The updated Preferred Operator PLMN list may have changed as a result of the secured packet. One will appreciate that the Preferred Operator PLMN list is an entry in the USIM application that can only

be modified by the entity that has the necessary keys that allow modifications to be performed e.g. Home service provider, the HPLMN. However it could be any application that provides the functionality to the ME to indicate which networks should be chosen when the UE performs a PLMN selection.

(20) In some cases, such as in 5G networks, an authentication procedure (which is part of a registration procedure) can be used to transport a list of networks or PLMN s (or a list of preferred PLMN and access technology combinations) to the UE. FIG. 1 is a data flow diagram **100** illustrating an example process of using an authentication procedure to transport a list of preferred PLMN and access technology combinations. The data flow diagram **100** is according to 3GPP CTI contribution Cl-180462 which is hereby incorporated by reference. The data flow diagram **100** includes a UE **102**, a VPLMN access and mobility management function (AMF) node **104**, an HPLMN authentication server function (AUSF) node **106**, and an HPLMN unified data management (UDM) node **108**. The UE **102** can include an ME and a (removable) memory module. An example of a memory module is a UICC. The UICC may include a subscriber identity module (SIM), a Universal SIM (USIM), or a Removable User Identity Module (RUIM) which are collectively known as UICC applications. The ME and the UICC can use a set of commands, USAT commands defined in 3GPP TS 31.111 to exchange data and request each other to perform operations on behalf of the other entity. AUE may also be known as a mobile station (MS). The nodes **104**, **106**, and **108** can be core network components. The AMF can include access and mobility management functions such as registration management, connection management, mobility management, and access authentication and authorization. The AUSF can provide UE authentication services. The UDM can have similar functionalities as Home Location Register (HLR) or Home Subscriber Server (HSS), such as generating 3GPP Authentication and Key Agreement (AKA) authentication credentials. In this disclosure, access technology refers to radio access technology.

(21) In FIG. 1, during steps 5-7 of the authentication procedure, the HPLMN UDM **108** can send SOR data (e.g., a list of preferred PLMN, a list of preferred PLMN and access technology combinations, or an HPLMN protected list of preferred PLMN/access technology combinations) to the UE **102**. For example, if the HPLMN wants the UE to register on a different PLMN (3.sup.rd network), the HPLMN UDM **108** can send the SOR data. However, after the UE **102** receives the SOR data, the UE cannot perform a PLMN search because the UE is in a connected mode during the authentication phase (as discussed above, the PLMN search is performed when the UE is in an idle mode). In some cases, the timer T is not set until the UE completes the registration procedure. In other words, the timer T is started upon completion of the registration procedure, and the UE does not perform a PLMN search until the timer T expires. As a result, the UE **102** can be registered on an undesirable PLMN for a considerable amount of time. However, it is beneficial for an operator to steer (request the UE to perform a PLMN search) the UE to a different PLMN at its earliest convenience, for example, before the attach or registration attempt successfully completes. Further, a UE in the authentication phase cannot use the existing procedures of 2G, 3G, or 4G systems to receive a secured packet that includes the SOR data, because a UE cannot receive an SM if it is not attached or registered to a network.

(22) Besides that, the UE may be on an undesirable PLMN for a long time, the SOR data delivery in FIG. 1 also has the following issues. First, in steps 5-7, the SOR data is sent in clear text to the UE **102**. As a result, the SOR data can be modified by the VPLMN node **104**. Second, the authentication procedure in 5G networks uses Extensible Authentication Protocol (EAP). In some cases, the data in the first EAP packet can be modified and even removed by the VPLMN node **104**, and the receiving UE **102** and the sending HPLMN node **108** do not know that this has been done. Third, persistent information on the USIM or UICC may not be updated, causing the HPLMN to possibly send SOR data repeatedly. A UICC or USIM (e.g., memory module) contains persistent data or information. Persistent data refers to data stored in a memory which would not be

rendered inaccessible or even wiped or cleared, upon, e.g., restart of the device or module within the device. For example, the ME reads, e.g., the data “Operator Controlled PLMN Selector with Access Technology” or other data from the USIM or UICC, e.g., after the ME boots or activated. The ME may read the data at other times, e.g., upon receiving the REFRESH command. The ME may modify the data “Operator Controlled PLMN Selector with Access Technology” based on the received SOR data, and the modification made by the ME should be synced to the USIM or UICC so that the persistent data is updated and the updates are available after, e.g., reboot, boot up or startup. However, write access to some UICC or USIM data, including the “Operator Controlled PLMN Selector with Access Technology”, is protected as described earlier. Only the HPLMN operator has the credentials for the write access to this data. Neither the VPLMN nor the ME has these credentials. In other words, after receiving the SOR data, the ME in the UE **102** cannot update the persistent data on the UICC or USIM, which may cause the HPLMN to send SOR data repeatedly.

(23) In some cases, a UE can be in a manual network selection mode. In the manual network selection mode, the UE selects a PLMN without necessarily considering the HPLMN's preferences. In an automatic network selection mode, the UE does consider the HPLMN's preferences when selecting a PLMN. In the case of manual network selection mode, SOR may not take place because the device has chosen a VPLMN or network per the UE's preferences. Another reason for a UE selecting a network or VPLMN is because the VPLMN is included in the “User Controlled PLMN Selector with Access Technology” list. For example, the UE finds a network on the “User Controlled PLMN Selector with Access Technology” list. The user/application has populated the “User Controlled PLMN Selector with Access Technology” list with PLMN entry(s) and it is akin to performing a manual network selection, i.e., the UE has chosen a network for a specific reason. Therefore, SOR may not take place if the UE has chosen a VPLMN and a network in the “User Controlled PLMN Selector with Access Technology” list.

(24) In some cases, a UE can be battery constrained (or resource constrained). Because the PLMN search (e.g., network discovery process) consumes battery power and there is no guarantee that an alternative network is available, it is desirable that SOR operations take into account if a UE is battery constrained to prolong battery life. In some cases, a UE can be mobility constrained such as fixed UEs, and it may be desirable not to perform SOR because the PLMNs available to the UE are not likely to change.

(25) The SOR procedure, according to methods and systems described herein, enables an operator to steer a UE to a different PLMN at the earliest convenience, e.g., before the attach or registration attempts successfully completes. The described approach uses a secured packet to deliver SOR data so that intermediate nodes along the path cannot modify the SOR data. The described approach can also update the persistent information on the UICC based on the SOR data. The described method also informs the PLMN if the UE will be unable to act on the SOR information due to the automatic network selection mode, user controlled PLMN selector list, or manual network selection mode. Another reason why the UE is unable to select a different PLMN is because the PLMN currently being registered continues to be the highest priority PLMN. Finally, the secured packet may fail an integrity check at the UICC. In any of these cases, the ME may be requested, via USAT, by the memory module to transmit a second secured packet to the network. The second secured packet may indicate to the network the reasons why the UE cannot select another PLMN or the integrity check failure. In some cases, a security check comprises an integrity check. In this disclosure, terms “security check” and “integrity check” can be interchangeable. Security check may also be determining by the ME or UICC that one to many information elements, indicators, SOR data or secured packet that should have been present (expected) in a received message are received in the received message. Determining could be based on configuration within the ME and or UICC.

(26) 5G terminologies used in this disclosure are described below.

- (27) 5G system mobility management (5GMM)-IDLE mode: The term is used standalone. A UE in 5GMM-IDLE mode means the UE can be either in 5GMM-IDLE mode over 3GPP access or in 5GMM-IDLE mode over non-3GPP access.
- (28) 5GMM-CONNECTED mode: The term is used standalone. A UE in 5GMM-CONNECTED mode means the UE can be either in 5GMM-CONNECTED mode over 3GPP access or in 5GMM-CONNECTED mode over non-3GPP access.
- (29) 5GMM-IDLE mode over 3GPP access: A UE is in 5GMM-IDLE mode over 3GPP access when no N1 non-access stratum (NAS) signaling connection between the UE and network over 3GPP access exists. The term 5GMM-IDLE mode over 3GPP access used in this disclosure corresponds to the term Connection Management IDLE (CM-IDLE) state for 3GPP access used in 3GPP TS 23.501.
- (30) 5GMM-CONNECTED mode over 3GPP access: A UE is in 5GMM-CONNECTED mode over 3GPP access when an N1 NAS signaling connection between the UE and network over 3GPP access exists. The term 5GMM-CONNECTED mode over 3GPP access used in the present document corresponds to the term CM-CONNECTED state for 3GPP access used in 3GPP TS 23.501.
- (31) 5GMM-IDLE mode over non-3GPP access: A UE is in 5GMM-IDLE mode over non-3GPP access when no N1 NAS signaling connection between the UE and network over non-3GPP access exists. The term 5GMM-IDLE mode over non-3GPP access used in this disclosure corresponds to the term CM-IDLE state for non-3GPP access used in 3GPP TS 23.501.
- (32) 5GMM-CONNECTED mode over non-3GPP access: A UE is in 5GMM-CONNECTED mode over non-3GPP access when it has an N1 NAS signaling connection between the UE and network over non-3GPP access exists. The term 5GMM-CONNECTED mode over non-3GPP access used in this disclosure corresponds to the term CM-CONNECTED state for non-3GPP access used in 3GPP TS 23.501.
- (33) Access stratum connection: A peer to peer access stratum connection between either the UE and the Next Generation-radio access network (NG-RAN) for 3GPP access or the UE and the N3IWF for non-3GPP access. The access stratum connection for 3GPP access corresponds to a radio resource control (RRC) connection via the Uu reference point. The creation of the access stratum connection for non-3GPP access corresponds to the completion of the IKE_SA_INIT exchange (see IETF RFC 7296) via the NWu reference point.
- (34) N1 NAS signaling connection: A peer to peer N1 mode connection between UE and AMF. An N1 NAS signaling connection is either the concatenation of an RRC connection via the Uu reference point and an NG connection via the N2 reference point for 3GPP access, or the concatenation of an IPsec tunnel via the NWu reference point and an NG connection via the N2 reference point for non-3GPP access.
- (35) 5G PLMN search procedure is described as follows.
- (36) The PLMN search procedure for 5G is substantially identical to PLMN search procedures for PLMNs using EPC or GPRS core networks. As discussed above, a PLMN search happens initially and a PLMN search also happens periodically. The PLMN search that happens periodically occurs when a timer (e.g., the timer T discussed above) expires. The periodically happening PLMN search searches for a higher priority PLMN.
- (37) For example, 3GPP TS 23.122 provides the following description for the PLMN search procedure. If the MS is in a VPLMN, the MS shall periodically attempt to obtain service on its HPLMN (if the EHPLMN list is not present or is empty) or one of its EHPLMNs (if the EHPLMN list is present) or a higher priority PLMN/access technology combinations listed in “user controlled PLMN selector” or “operator controlled PLMN selector” by scanning in accordance with the requirements that are applicable to i), ii) and iii) as below. In the case that the mobile has a stored “Equivalent PLMNs” list the mobile shall only select a PLMN if it is of a higher priority than those of the same country as the current serving PLMN which are stored in the “Equivalent PLMNs” list.

For this purpose, a value of timer T may be stored in the SIM. The interpretation of the stored value depends on the radio capabilities supported by the MS. The MS selects and attempts registration on other PLMN/access technology combinations, if available and allowable, in the following order: i) either the HPLMN (if the EHPLMN list is not present or is empty) or the highest priority EHPLMN that is available (if the EHPLMN list is present); ii) each PLMN/access technology combination in the “User Controlled PLMN Selector with Access Technology” data file in the SIM (in priority order); iii) each PLMN/access technology combination in the “Operator Controlled PLMN Selector with Access Technology” data file in the SIM (in priority order).

(38) In addition to using the “Operator Controlled PLMN Selector” list, the device or UE can also use the “User Controlled PLMN Selector with Access Technology” list. If the device, when performing a PLMN search finds a PLMN that is in the “User Controlled PLMN Selector with Access Technology”, then the device will choose this PLMN (with higher priority) than any PLMN in “Operator Controlled PLMN Selector”.

(39) Turning to a general description of the elements, a UE may be referred to as a mobile electronic device, user device, mobile station, subscriber station, portable electronic device, mobile communications device, wireless modem, wireless terminal, mobile equipment, session initiation protocol (SIP) user agent, set-top box, test equipment, or embedded modem. Examples of a UE may include a cellular phone, personal data assistant (PDA), smart phone, laptop, tablet personal computer (PC), pager, portable computer, portable gaming device, wearable electronic device, or other mobile communications device having components for communicating data via a wireless communication network. The wireless communication network may include a wireless link over at least one of a licensed spectrum and an unlicensed spectrum.

(40) Other examples of a UE include mobile and fixed electronic devices. A UE may include a ME device and a removable memory module, such as a UICC that includes a SIM application, a USIM application, or an R-UIM application all known as UICC applications. The term “UE” can also refer to any hardware or software component that can terminate a communication session for a user. In addition, the terms “user equipment,” “UE,” “user equipment device,” “user agent,” “UA,” “user device,” and “mobile device” can be used synonymously herein. A UICC could also be a secure element that contains UICC applications that perform similar functionality.

(41) The wireless communication network may include one or a plurality of radio access networks (RANs), other access networks such as fixed Ethernet or IEEE 802.11 WLAN, core networks (CNs), and external networks. The RANs may comprise one or more radio access technologies. The radio access technologies can be 3GPP access technologies or non-3GPP access technologies. In some implementations, the radio access technologies may be Global System for Mobile communication (GSM), Interim Standard 95 (IS-95), Universal Mobile Telecommunications System (UMTS), CDMA2000 (Code Division Multiple Access), Evolved Universal Mobile Telecommunications System (UMTS), Long Term Evolution (LTE), LTEAdvanced, or 5G access technologies. In some instances, the core networks may be evolved packet cores (EPCs) or 5G cores. The core networks may include AMF, Session Management Function (SMF), UDM, Authentication, Authorization, and Accounting (AAA) server, or other network nodes or entities.

(42) FIG. 2 is a data flow diagram 200 illustrating an example SOR procedure, according to some implementations of the present disclosure. The data flow diagram 200 includes a UE having a UICC 202 and a ME 204, a first network node 206, a second network node 208, and a third network node 210. The first network node 206 can be an AMF or SMF node of a first VPLMN (2.sup.nd network), e.g., PLMNa. The second network node 208 can be an AMF or SMF node of a second VPLMN (3.sup.rd network), e.g., PLMNb. The third network node 210 can be a UDM node of an HPLMN (1.sup.st network).

(43) At step 1, the ME 204 sends a message to the first network node 206 to start a registration/attach procedure with PLMNa. For example, the ME 204 can send a REGISTRATION REQUEST message to the first network node 206. In some cases, an N1 NAS signaling connection

between the ME **204** and the first network node **206** can carry the REGISTRATION REQUEST message. The UE can use 3GPP access or non-3GPP access technologies. The N1 NAS signaling connection can be over 3GPP access or non-3GPP access. For example, the UE can be in a 5GMM-CONNECTED mode over 3GPP access, and an N1 NAS signaling connection over 3GPP access exists between the UE and first network node **206**.

(44) At step **1a**, the first network node **206** forwards the message in step **1** (e.g., the REGISTRATION REQUEST) to the third network node **210** in HPLMN. The first network node **206** can forward the message to the third network node **210** without going through the second network node **208**. In some cases, as will be discussed in FIGS. **3A-5**, the messages of steps **1** and **1a** can include an indication that indicates the type of device (e.g., battery constrained or mobility constrained) and the operating mode (e.g., manual network selection mode). This indication can help the third network node **210** in HPLMN to determine whether to send SOR data to the UE. For example, if the UE is battery constrained, mobility constrained, or in a manual network selection mode, a node within the HPLMN (e.g. third network node) may not send SOR data to the UE.

(45) At step **2**, the third network node **210** sends a secured packet to the first network node **206**. The secured packet can include SOR data. At step **3**, as will be discussed in FIGS. **6-8E**, the first network node **206** sends the secured packet received in step **2** to the ME **204**. In other words, the secured packet is received during the registration/attach procedure. In some cases, in step **3**, the secured packet is received in a DL NAS TRANSPORT message. In some other implementations, step **3** can be a REGISTRATION ACCEPT message or an ATTACH ACCEPT message including the SOR data or the secured packet. In some cases, EAP is used for the authentication procedure in the registration procedure (e.g., 5G networks use EAP for authentication), and the secured packet can be received in an EAP message. In some cases, the secured packet can be packets in a SM, and hence “secured packet” can be interchanged with “SM containing a secured packet”.

(46) At step **4**, the ME **204** determines by an indication (e.g., code point) in the message received at step **3** that the content is for the UICC, and the ME **204** sends the secured packet to the UICC **202**. The UICC **202** decodes the secured packet and retrieves the SOR data. In some cases, during steps **2-4**, the secured packet is encoded by the third network node **210** in HPLMN and decoded by the UICC **202**, while the intermediate entities such as the first network node **206** and the ME **204** do not decode the secured packet. Based on the SOR data, the UICC **202** can make decisions whether to trigger PLMN search.

(47) At step **5**, the ME **204** receives an indication from the UICC **202**. The indication can indicate the ME **204** to terminate the ongoing registration/attach procedure and trigger a PLMN search. In some cases, the indication from the UICC **202** can be received via a USAT command. The USAT command can be a REFRESH command. The REFRESH command can include an indication that an SOR procedure should be initiated. In some cases, as will be discussed in FIG. **9**, the REFRESH command at step **5** can optionally contain a list of preferred PLMNs so that the ME **204** can perform the PLMN search based on the list. In some cases, the REFRESH command does not contain the list of preferred PLMNs, and the REFRESH command triggers the ME **204** to download an environment file from the UICC **202** that includes a list of preferred PLMNs. The ME **204** then performs the PLMN search based on the downloaded list.

(48) At step **6**, based on the indication in step **5**, the ME **204** can terminate the registration/attach procedure with PLMN_a. In some cases, for terminating the registration/attach procedure, the ME **204** can release the N1 NAS signaling connection between the ME **204** and the first network node **206**. In some cases, the termination of the registration/attach procedure includes the ME **204** sending an authentication failure message or a REGISTRATION COMPLETE message to the first network node **206**. The authentication failure message or the REGISTRATION COMPLETE message can include an indicator, either indicating a failed receipt of the secured packet (e.g., the secured packet failed an integrity check as discussed below), a successful receipt of the secured packet, or indicating that the ME will not perform a PLMN search with an optional qualification

for the reason why. The indicator indicating failure can prevent the network (e.g., PLMN_a) from retransmitting a registration procedure related message. The indication indicating failure could be further qualified to indicate the actual reason, examples being but not limited to: PLMN search, PLMN temporary not allowed etc. When the indicator indicates a successful receipt, the indicator identifies to the network that the secured packet was successfully received and the ME will perform a PLMN search. When the indicator indicates that the ME will not perform a PLMN search, the indicator may be qualified indicating the reason: ME is in a manual network selection mode, VPLMN (RPLMN) is on the User controlled PLMN list, there are no other available PLMNs, PLMN temporary not allowed etc.

(49) At step 7, the ME **204** can start the PLMN search to find an alternative network (e.g., PLMN_b) to attach or register. At step 8, the ME **204** starts the registration/attach procedure with PLMN_b.

(50) In some cases, after the ME **204** receives the USAT command in step 5, if the ME is a device type of at least one of but not limited to battery constrained, resource constrained, mobility constrained, or the ME has selected a network (RPLMN) because that network was in the User controlled PLMN list (i.e., that network is a user-preferred PLMN), the ME **204** does not perform the PLMN search until when either the periodical PLMN search timer T expires or when the ME **204** performs a PLMN search for other reasons. In some cases, when the ME **204** receives the USAT command in step 5, if the UE is in a manual network selection mode or the ME has selected a network (RPLMN) because that network was in the User controlled PLMN list, the UE refrains from performing a PLMN search. The User controlled PLMN list is also called “User Controlled PLMN Selector with Access Technology” list. In some cases, the ME **204** can inform the network (e.g., a node of the HPLMN (e.g., third network or third network's node)) that it is in a manual network selection mode or the ME has selected a network (RPLMN) because that network was in the User controlled PLMN list so that a node of the HPLMN (third network node) does not send SOR data.

(51) In some cases, the indication received from the UICC **202** in step 5 indicates that the secured packet failed an integrity check at the UICC. The indication that the secured packet failed the integrity check may cause the ME **204** to remove the PLMN with which the UE is attempting to register (e.g., PLMN_a) from the “Operator Controlled PLMN Selector with Access Technology” list stored in the ME **204** and optionally include PLMN_a in a forbidden PLMN list (e.g., EF.sub.FPLMN(Forbidden PLMNs)). In some cases, if the current registration/attach procedure with PLMN_a is for emergency services, the UE starts the PLMN search after there is no longer a need for emergency services. In some implementations, in step 6, the ME may send a message, e.g., a REGISTER COMPLETE message or an ATTACH COMPLETE message, to the network containing an indication indicating that the secured packet failed the integrity/security check, that the VPLMN is on the User controlled PLMN list or that the ME is in a manual network selection mode. The sending of the message maybe dependent on the ME being configured to respond (e.g., the ME is configured to respond the REGISTRATION ACCEPT/ATTACH ACCEPT message in step 3). This configuration may be stored in an Open Mobile Alliance (OMA) device management (DM) file in the ME, read from the UICC and then stored in the ME, or could have been received in the REGISTRATION ACCEPT/ATTACH ACCEPT message in step 3. In some cases, the ME is configured to expect to receive SOR data in the REGISTRATION ACCEPT/ATTACH ACCEPT message in step 3 e.g. either as indication in the REGISTRATION ACCEPT/ATTACH ACCEPT message in step 3 or as an indication in the SOR data or secured packet that was in the REGISTRATION ACCEPT/ATTACH ACCEPT message in step 3. This configuration may be stored in the OMA DM file in the ME, or read from the UICC and then stored in the ME.

(52) In some case, if the UE is in a manual network selection mode, the following procedure can be performed: 1. The ME starts registration/attach procedure with a VPLMN. 2. The ME receives a first secured packet including SOR data from, e.g., UDM in HPLMN. 3. The UE determines that it is in a manual network selection mode. 4. The ME sends the first secured packet to the UICC along

with an indication that the UE is in a manual network selection mode. 5. The ME receives a second secured packet from the UICC, where the second secured packet includes an indication that the ME is in a manual network selection mode. 6. The ME can optionally send to the network the second secured packet (e.g., HPLMN) and optionally include an indication that the ME is in a manual network selection mode so that the HPLMN does not send SOR data. In some cases, the ME can continue the current registration/attach procedure with the VPLMN.

(53) In some cases, if the UE is in a manual network selection mode or the UE selected a PLMN that was in the User controlled PLMN list, the following procedure can be performed: 1. The ME starts registration/attach procedure with a VPLMN. 2. The ME receives a secured packet including SOR data from, e.g., UDM in HPLMN. For example, the secured packet can be received using EAP as shown in FIGS. 6-8E. 3. The ME sends the secured packet to the UICC. 4. The ME receives an indication from the UICC to perform a PLMN search. For example, the USAT REFRESH command from the UICC can include an indicator to perform a PLMN search. 5. Because the ME is in a manual network selection mode or the UE selected a PLMN that was in the User controlled PLMN list, the ME decides not to perform the PLMN search and continues the current registration/attach procedure with the VPLMN. For example, the ME can send an EAP response message to the network, where the EAP message, e.g., EAP-response, may contain an indication why the ME is continuing with the PLMN search, e.g., in the manual network selection mode, or UE used the User controlled PLMN list.

(54) In this disclosure, the described approach for the case when the UE is in a manual network selection mode is also applicable to the case when the ME selected a network (PLMN) in the “User Controlled PLMN Selector with Access Technology” list.

(55) As discussed above, in step 1 of FIG. 2 the ME can indicate the device type (e.g., battery constrained, mobility constrained device, and in a manual network selection mode). FIGS. 4A-5 describes two methods for indicating the device type. FIGS. 3A-3B illustrate a data flow diagram 300 of an example registration procedure from 3GPP TS 23.502. FIGS. 4A-4C illustrate including the device type indicator in a REGISTRATION REQUEST message (e.g., step 1 of FIG. 3A), while FIG. 5 illustrates including the device type indicator in an EAP message (i.e., step 9 of FIG. 3A).

(56) FIGS. 4A-4C illustrate an example description for including a device type indicator in a REGISTRATION REQUEST message, according to some implementations of the present disclosure. For example, the registration initiation procedure described in 3GPP TS 24.501 can be modified to include the underlined text shown in FIGS. 4A-4C. Table 8.2.5.1.1 in FIG. 4B and Table 9.8.2.2.1 in FIG. 4C illustrate that the REGISTRATION REQUEST message can include a new information element “device type” to indicate whether the UE is a battery or resource constrained device, a mobility constrained device, and/or in a manual network selection mode. In some implementations, the setting of the battery constrained indicator can change if the device is connected to a power supply, as such the ME might send a mobility management message including the device type. One skilled in the art will appreciate that in this disclosure message names, code point names, etc. are used for illustrative purposes, and other message names and code point names can be used. For example, indicators can be sent using new information elements or extending existing information elements. In this disclosure, occurrences of “shall” could be “may” or “should”.

(57) FIG. 5 illustrates an example description for including a device type indicator in an EAP message, according to some implementations of the present disclosure. For example, 3GPP TS 24.302 can be modified to include the underlined text shown in FIG. 5. Table 8.2.X. 1-1 in FIG. 5 shows that an EAP-Response/AKA'-Challenge message can include an AT_SORInfo_Request attribute containing the device type indicator.

(58) As discussed above, in step 2-3 of FIG. 2 the HPLMN (e.g., UDM) can send the secured packet including SOR data to the UE. FIGS. 6-8E describe methods for sending the SOR data.

(59) FIG. 6 is a data flow diagram **600** illustrating an example procedure using EAP signaling to send SOR data, according to some implementations of the present disclosure. The data flow diagram **600** includes a UE **602**, an AAA server **604** in a VPLMN, and a database or UDM **606** in HPLMN. The AAA server **604** can also be replaced with AMF and/or UDM. At step **4**, the AAA server **604** in VPLMN sends an authentication challenge to the UE **602**. The authentication challenge can include an AT_SORInfo_Request_Supported attribute (described in section 8.2.X.1 in FIG. **8C**) indicating that the AAA server **604** supports the UE **602** requesting SOR data. At step **5**, in response to receiving the indicator that the AAA server **604** supports the UE **602** requesting SOR data, the UE **602** sends an authentication response to the AAA server **604**, where the authentication response includes an AT_SORInfo_Request attribute (described in section 8.2.X.2 in FIG. **8E**) indicating that the UE **602** requests SOR data, optionally if the UE is battery constrained or operating in manual network selection mode. At step **6**, the AAA server **604** forwards the authentication response to the database **606** in HPLMN. The authentication response in step **6** also includes the AT_SORInfo_Request attribute indicating that the UE **602** requests SOR data. The UDM/HSS/HLR takes into account the operating mode of the UE and if it is battery constrained. At step **7**, in response to receiving the indicator indicating that the UE **602** requests SOR data, the database **606** in HPLMN sends an authentication acknowledgement to the AAA server **604**, where the authentication acknowledgement includes an AT_SORInfo_RESP attribute (described in section 8.2.X.3 in FIGS. **8D-8E**) that contains the SOR data. At step **8**, the AAA server **604** forwards the authentication acknowledgement to the UE **602**. The authentication acknowledgement in step **8** also includes the AT_SORInfo_RESP attribute containing the SOR data.

(60) FIG. 7 is a data flow diagram **700** illustrating an example process using EAP signaling for obtaining SOR data in 5G networks, according to some implementations of the present disclosure. The example process is also applicable to other EAP framework methods, e.g., EAP methods used to access wireless local area networks (WLANs), where the names of the functions can be different. The data flow diagram **700** includes a UE **702**, an Security Anchor Functionality (SEAF) or AMF node **704** in a VPLMN, a Authentication Server Function (AUSF) node **706** in an HPLMN, and a UDM/Authentication credential Repository and Processing Function (ARPF) node **708** in the HPLMN. Similar to FIG. **6**, at step **4** of FIG. **7**, the EAP-Request/AKA'-Challenge can include the AT_SORInfo_Request_Supported attribute indicating that the SEAF/AMF **704** supports the UE **702** requesting SOR data. At steps **5** and **6** of FIG. **7**, the EAP-Response/AKA'Challenge can include the AT_SORInfo_Request attribute indicating that the UE **702** requests SOR data. The messages in steps **7** and **8** of FIG. **7** can include the AT_SORInfo_RESP attribute containing the SOR data. In some cases, a decision to send an indication that SOR is supported in step **2** and/or **3** is based upon the network and/or the location that the UE has requested to register on. In some cases, the data sent in step **7** could be sent in step **3**.

(61) FIGS. **8A-8E** illustrate an example description of sending SOR data in EAP-AKA', according to some implementations of the present disclosure. For example, 3GPP TS 24.302 can be modified to include the underlined text shown in FIG. **8**.

(62) In some cases, at step **5** in FIG. **2**, the ME can receive a USAT REFRESH command from the UICC. The REFRESH command can optionally contain a list of PLMNs for 5G access technology (e.g. NG or E-UTRAN connected to 5G core network), or a PLMN-with-access technology (PLMNwAct) list containing access technology selector including radio access technologies (RATs) used to determine to perform the 5G SOR procedure or steering of a UE from one VPLMN to another VPLMN.

(63) FIG. **9** illustrates an example description of a REFRESH command, according to some implementations of the present disclosure. For example, 3GPP TS 31.111 can be modified to include the underlined text shown in FIG. **9** so that the REFRESH command includes two new parameters “(5G) PLMN List” and “(5G) PLMNwAct list”. In some cases, the PLMN list and the PLMNwAct list can define the preferred PLMNs in a priority order. The ME can obtain the PLMN

list or the PLMNwActList from the REFRESH command, and performs the PLMN search based on the list, e.g., starting from the highest priority PLMN.

(64) FIG. 10 illustrates an example description of an environment file (EF) for data “Operator Controlled PLMN Selector with Access Technology”, according to some implementations of the present disclosure. For example, 3GPP TS 31.102 can be modified as shown in FIG. 10 to indicate three different embodiments, i.e., 5G System (5GS) supported by New Radio (NR) access, 5G System (5GS) supported by E-UTRA, or EPS (EPC supported by E-UTRA). In some cases, the EF file EF.sub.OPLMNwACT is on the UICC and includes preferred PLMNs in a priority order. The REFRESH command can trigger the ME to download EFEF.sub.OPLMNwACT to the memory of the ME so that the ME can perform the PLMN search based on the preferred PLMNs, e.g., starting from the highest priority PLMN.

(65) In some cases, the 5G SOR procedure or steering of a UE from one VPLMN to another VPLMN may involve terminating the ongoing registration procedure by at least one of a REGISTRATION COMPLETE message, an authentication failure message, or releasing the N1 NAS signaling connection.

(66) In some cases, upon the UE (or ME, UE and ME can be interchangeable in this disclosure) terminating the ongoing registration procedure, the UE continues operations as if the UE was switched on or the UE was recovered from lack of coverage, and the UE selects the highest priority PLMN that is available. Alternatively, upon the UE terminating the ongoing registration procedure, the UE selects the highest priority PLMN or equivalent highest priority PLMN (if it is available) using all access technologies via which it had previously discovered the previously highest priority PLMN. Upon failing to discover a higher priority PLMN than the previously highest priority PLMN, the UE using all access technologies that the UE is capable of and if necessary to discover a higher priority PLMN.

(67) As an alternative, if the UE is either battery constrained or the network was selected because the UE is operating in manual network selection mode or the UE had selected a PLMN from the user controlled PLMN list, the ME can ignore the REFRESH command but use the updated Preferred Operator PLMN list that has been read into the ME's memory when the ME performs a PLMN search, e.g., because the periodical search timer T expired. The ME continues operations as if the ME was switched on or the ME has lost PLMN coverage.

(68) FIG. 11 is a data flow diagram 1100 illustrating an example SOR procedure, according to some implementations of the present disclosure. The data flow diagram 1100 modifies the procedure in FIG. 1 based on the described approaches in the present disclosure. FIGS. 12A-12B illustrate an example description for the SOR procedure of FIG. 11, according to some implementations of the present disclosure. FIGS. 12A-12B show changes to the 3GPP CTI contribution C 1-180462.

(69) FIGS. 15A-15F illustrate an example description for steering of UE in VPLMN during registration and after registration, according to some implementations of the present disclosure. The example description in FIGS. 15A-15F can be included in 3GPP TS 23.122. Message 6 in figure C.1.1 of FIGS. 15A-15B (i.e., REGISTRATION ACCEPT) can be in step 3 of FIG. 2. Message 10 in figure C.1.1 of FIGS. 15A-15B (i.e., REGISTRATION COMPLETE) can be in step 6 of FIG. 2.

(70) In some cases, if the SOR data (e.g., the secured packet including the HPLMN protected list of preferred PLMN/access technology combinations) is successfully received (e.g., successful security check), and if the VPLMN the ME is currently attempting to register is not a user-preferred PLMN and the ME is not in a manual selection mode, the ME may terminate the current registration procedure and perform a PLMN search based on the SOR data e.g. after completion of the REGISTRATION/ATTACH procedure (step 10) or before in any of steps 7-9. In some cases, if the SOR data is successfully received, and if the VPLMN the ME is currently attempting to register is a user-preferred PLMN or the ME is in a manual selection mode, the ME may continue the current registration procedure and not to perform the PLMN search. In some cases, if the SOR data is not

successfully received (e.g., fails security check, or the ME is configured to receive the SOR data but did not receive), and if the VPLMN the ME is currently attempting to register is not a user-preferred PLMN and the ME is not in a manual selection mode, the ME may terminate the current registration procedure and perform a PLMN search. In some cases, if the SOR data is not successfully received, and if the VPLMN the ME is currently attempting to register is a user-preferred PLMN or the ME is in a manual selection mode, the ME may continue the current registration procedure and not to perform the PLMN search.

(71) In message 10 of figure C.1.1 in FIGS. **15A-15B** (i.e., REGISTRATION COMPLETE), the ME, if configured, can send one or more indications indicating that the SOR data was not received, that the SOR data failed security check, that the VPLMN the ME is currently attempting to register is a user-preferred PLMN, or that the ME is in a manual selection mode. As shown in the text in FIGS. **15C-15E**, there can be two options for operations associated with message 10. In the second option, operation 10a can be performed if the SOR data is not successfully received (e.g., unsuccessful or failed security check), and operation 10b can be performed if the SOR data is successfully received (e.g., successful security check).

(72) FIG. **13** is a schematic illustrating an example network node **1300** according to some implementations of the present disclosure. For example, the network nodes **206, 208, 210, 604, 606, 704, 706, and 708** can be implemented by the network node **1300**. The illustrated device **1300** includes a processing module **1302**, a wired communication subsystem **1304**, and a wireless communication subsystem **1306**. The wireless communication subsystem **1306** can receive data traffic and control traffic from the UE. In some implementations, the wireless communication subsystem **1306** may include a receiver and a transmitter. The wired communication subsystem **1304** can be configured to transmit and receive control information between other access node devices via backhaul connections. The processing module **1302** can include one or more processing components (alternatively referred to as “processors” or “central processing units” (CPUs)) capable of executing instructions related to one or more of the processes, steps, or actions described above in connection with one or more of the implementations disclosed herein. The processing module **1302** can also include other auxiliary components, such as random access memory (RAM), read only memory (ROM), secondary storage (for example, a hard disk drive, flash memory or other non-transitory storage medium). The processing module **1302** can execute certain instructions and commands to provide wireless or wired communication, using the wired communication subsystem **1304** or a wireless communication subsystem **1306**. Various other components can also be included in the device **1300**.

(73) FIG. **14** is a schematic illustrating an example UE **1400** apparatus according to some implementations of the present disclosure. The example UE **1400** includes a processing unit **1402**, a computer-readable storage medium **1404** (for example, ROM or flash memory), a wireless communication subsystem **1406**, an interface **1408**, and an I/O interface **1410**. The processing unit **1402** can include one or more processing components (alternatively referred to as “processors” or “central processing units” (CPUs)) configured to execute instructions related to one or more of the processes, steps, or actions described above in connection with one or more of the implementations disclosed herein. The processing unit **1402** can also include other auxiliary components, such as random access memory (RAM) and read only memory (ROM). The computer-readable storage medium **1404** can be embodied by a non-transitory medium configured to store an operating system (OS) of the device **1400** and various other computer-executable software programs for performing one or more of the processes, steps, or actions described above.

(74) The wireless communication subsystem **1406** may be configured to provide wireless communications for data information or control information provided by the processing unit **1402**. The wireless communication subsystem **1406** can include, for example, one or more antennas, a receiver, a transmitter, a local oscillator, a mixer, and a digital signal processing (DSP) unit. In some implementations, the subsystem **1406** can support multiple input multiple output (MIMO)

transmissions. In some implementations, the receivers in the wireless communication subsystems **1406** can be an advance receiver or a baseline receiver. Two receivers can be implemented with identical, similar, or different receiver processing algorithms.

(75) The user interface **1408** can include, for example, one or more of a screen or touch screen (for example, a liquid crystal display (LCD), a light emitting display (LED), an organic light emitting display (OLED), a microelectromechanical system (MEMS) display), a keyboard or keypad, a trackball, a speaker, and a microphone. The I/O interface **1410** can include, for example, a universal serial bus (USB) interface. A skilled artisan will readily appreciate that various other components can also be included in the example UE device **1400**.

(76) While operations are depicted in the drawings in a particular order, this should not be understood as requiring that such operations be performed in the particular order shown or in sequential order, or that all illustrated operations be performed, to achieve desirable results. In certain circumstances, multitasking and parallel processing may be employed. Moreover, the separation of various system components in the implementation described above should not be understood as requiring such separation in all implementations, and it should be understood that the described program components and systems can generally be integrated together in a signal software product or packaged into multiple software products.

(77) Also, techniques, systems, subsystems, and methods described and illustrated in the various implementations as discrete or separate may be combined or integrated with other systems, modules, techniques, or methods. Other items shown or discussed as coupled or directly coupled or communicating with each other may be indirectly coupled or communicating through some interface, device, or intermediate component, whether electrically, mechanically, or otherwise. Other examples of changes, substitutions, and alterations are ascertainable by one skilled in the art and may be made.

(78) While the above detailed description has shown, described, and pointed out the fundamental novel features of the disclosure as applied to various implementations, it will be understood that various omissions, substitutions, and changes in the form and details of the system illustrated may be made by those skilled in the art. In addition, the order of method steps is not implied by the order they appear in the claims.

Claims

1. A user equipment, UE, comprising a processor configured to: send a registration request message to a visited public land mobile network, VPLMN; receive a registration accept message from the VPLMN, the registration accept message comprising steering of roaming information received from a home public land mobile network, HPLMN; determine whether the steering of roaming information includes a list of preferred public land mobile network, PLMN/access technology combinations and whether a security check of the steering of roaming information is successful; based on determining that the steering of roaming information includes the list of preferred PLMN/access technology combinations and that a security check of the steering of roaming information is successful: replace a highest priority entry in an operator controlled PLMN selector with access technology list with the received list of preferred PLMN/access technology combinations, and send a registration complete message to the VPLMN, the registration complete message comprising a transparent container including a UE response to the HPLMN; and add the VPLMN to a list of forbidden PLMNs based on determining that the security check of the steering of roaming information fails.

2. The UE of claim 1, wherein the processor is further configured to send the registration complete message on condition that the security check of the steering of roaming information fails or the list of preferred PLMN/access technology combinations is not contained in the steering of roaming information, the registration complete message comprising an indication that the security check

failed.

3. The UE of claim 1, wherein a unified data management (UDM) node requests a response to the response to the registration accept message from the UE.

4. The UE of claim 1, wherein the UE includes a Universal Integrated Circuit Card, UICC.

5. The UE of claim 4, wherein the steering of roaming information is included in a secured packet.

6. The UE of claim 5, wherein the processor is configured to send the secured packet to the UICC.

7. The UE of claim 6, wherein the processor is configured to receive an indication regarding whether the security check of the steering of roaming information was successful from the UICC.

8. The UE of claim 7, wherein the indication is comprised in a USAT command.

9. A method implemented by a user equipment, UE, the method comprising: sending a registration request message to a visited public land mobile network, VPLMN; receiving a registration accept message from the VPLMN, the registration accept message comprising steering of roaming information received from a home public land mobile network, HPLMN; determining whether the steering of roaming information includes a list of preferred public land mobile network, PLMN/access technology combinations and whether a security check of the steering of roaming information is successful; based on determining that the steering of roaming information includes the list of preferred PLMN/access technology combinations and that a security check of the steering of roaming information is successful: replacing a highest priority entry in an operator controlled PLMN selector with access technology list with the received list of preferred PLMN/access technology combinations, and sending a registration complete message to the VPLMN, the registration complete message comprising a transparent container including a UE response to the HPLMN; and adding the VPLMN to a list of forbidden PLMNs based on determining that the security check of the steering of roaming information fails.

10. The method of claim 9, further comprising: sending the registration complete message based on determining that the security check of the steering of roaming information fails or the list of preferred PLMN/access technology combinations is not contained in the steering of roaming information, the registration complete message comprising an indication that the security check failed.

11. The method of claim 9, wherein a unified data management (UDM) node requests a response to the response to the registration accept message from the UE.

12. The method of claim 9, wherein the UE includes a Universal Integrated Circuit Card, UICC.

13. The method of claim 12, wherein the steering of roaming information is included in a secured packet.

14. The method of claim 13, wherein the UICC receives the secured packet.

15. The method of claim 14, further comprising receiving from the UICC an indication regarding whether the security check of the steering of roaming information was successful.

16. The method of claim 9, wherein the VPLMN added to the list of forbidden PLMNs is an undesirable PLMN.

17. A non-transitory machine-readable medium comprising instructions that, upon execution, instruct a user equipment, UE, to: send a registration request message to a visited public land mobile network, VPLMN; receive a registration accept message from the VPLMN, the registration accept message comprising steering of roaming information received from a home public land mobile network, HPLMN; determine whether the steering of roaming information includes a list of preferred public land mobile network, PLMN/access technology combinations and whether a security check of the steering of roaming information is successful; based on determining that the steering of roaming information includes the list of preferred PLMN/access technology combinations and that a security check of the steering of roaming information is successful: replace a highest priority entry in an operator controlled PLMN selector with access technology list with the received list of preferred PLMN/access technology combinations, and send a registration complete message to the VPLMN, the registration complete message comprising a transparent

container including a UE response to the HPLMN; and add the VPLMN to a list of forbidden PLMNs based on determining that the security check of the steering of roaming information fails.

18. The non-transitory machine-readable storage medium of claim 17, wherein the instructions further instruct the UE to: send the registration complete message based on determining that the security check of the steering of roaming information fails or the list of preferred PLMN/access technology combinations is not contained in the steering of roaming information, the registration complete message comprising an indication that the security check failed.

19. The non-transitory machine-readable storage medium of claim 17, wherein a unified data management (UDM) node requests a response to the response to the registration accept message from the UE.

20. The non-transitory machine-readable storage medium of claim 17, wherein the UE includes a Universal Integrated Circuit Card, UICC and the steering of roaming information is included in a secured packet, wherein the UE is instructed to send the secured packet to the UICC, and the UICC provides an indication regarding whether the security check of the steering of roaming information was successful to the UE.
